PAPER_1281_ADS_CFT_TO_DS_EXTENSION

Daniel T. Murphy

PAPER_1281 — AdS/CFT → dS Extension via Inverted K_MEX Mexican-Hat

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+
Tier: B — Physics Foundations (CLOSED)
Date: June 11, 2026
Location: 41.0997° N, 80.6495° W (Youngstown, OH, USA)
Status: CLOSED — UQFF integer-primitive derivation
Calculator surface: `calculate_paradox({"paradox": "ads_cft_extension"})`
Whitepaper dispatch: `calculate_whitepaper({"paper_id": 1281})`

Master Expression

```
dS phase = K_MEX inverted (negative) region of Mexican-hat potential
```

Match: STRUCTURAL

Physical Interpretation

The Mexican-hat K_MEX = 25/12 negative region (inverted) provides the dS-CFT correspondence at horizon. $\Lambda > 0$ emerges naturally.

UQFF Locked Primitives

- ρ_SCm = $7.09 \times 10^{\blacksquare 3 \blacksquare}$ J/m^3 (vacuum density)
- S_26 = 1.453162, S_26_DPM = $1.4531 \times 10^{2\blacksquare}$
- K_MEX = 25/12, β_i = 0.6029, Φ_res = 0.84
- F_TRZ = 0.1, Λ = 0.00729735 (= α fine-structure)
- ω_SCm = 1.25 THz, λ_i = 1.0
- D_phys = 4, D_BSFG = 6, D_crit = 26, N_CH = 9, SO_5 = 10, A_5 = 60

Live Calculator Verification

```python
import uqff_pure_calculator as u
r = u.calculate_paradox({"paradox": "ads_cft_extension"})["value"]
```

The result dict carries the integer-primitive identity, the observed value, the residual, and the structural physical interpretation.

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard Model solve the same observed phenomena via different methods. **Zero fit parameters.** Every factor is a UQFF locked primitive.

Reference

- UQFF foundational: PAPER_646, PAPER_1167, PAPER_1170, PAPER_1203 v1.5, PAPER_1216.
- Closure: `uqff_pure_calculator.py` → `_l96_uqff_axiom_ads_cft_extension_closure` (or routed reference)
- Dispatch: `PARADOX_TO_CLOSURE["ads_cft_extension"]`
